



U.S. National
Science Foundation



U.S. DEPARTMENT
of **ENERGY** | Office of
Science

Rubin 2036

NOIRLab and Rubin

Bob Blum



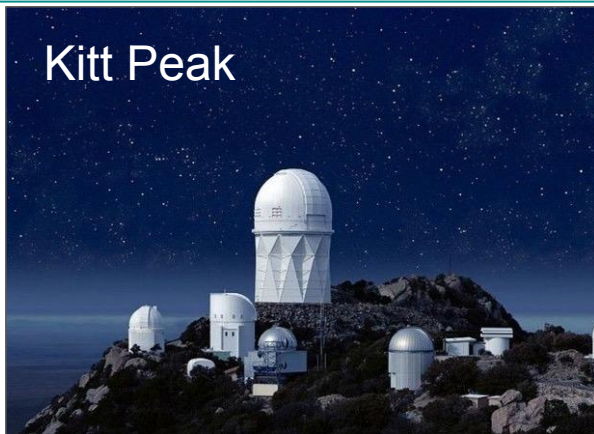
Thanks to Phil Marshall
for useful conversations

Rubin 2036

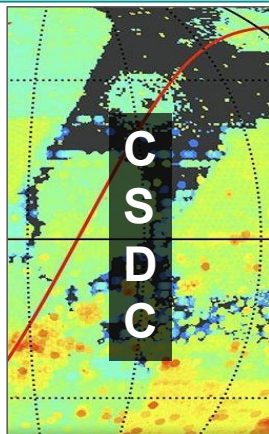
- NOIRLab's perspective on plans and processes for the future
- Potential operational / programmatic constraints or challenges for Rubin

NSF NOIRLab

Kitt Peak

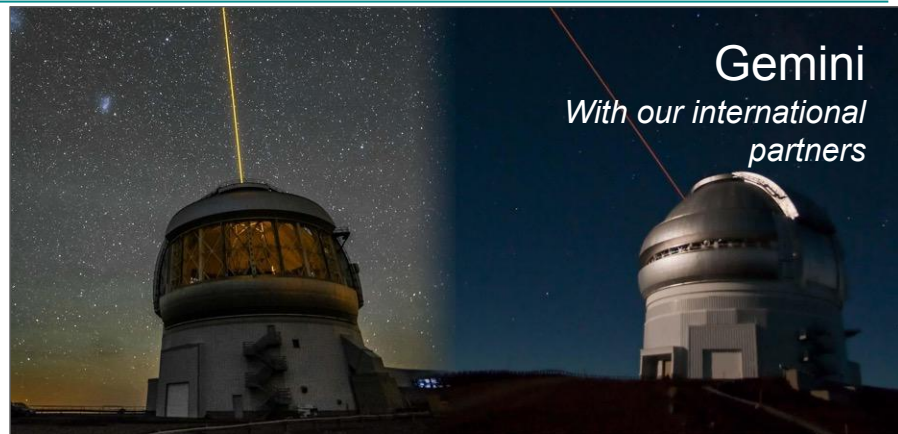


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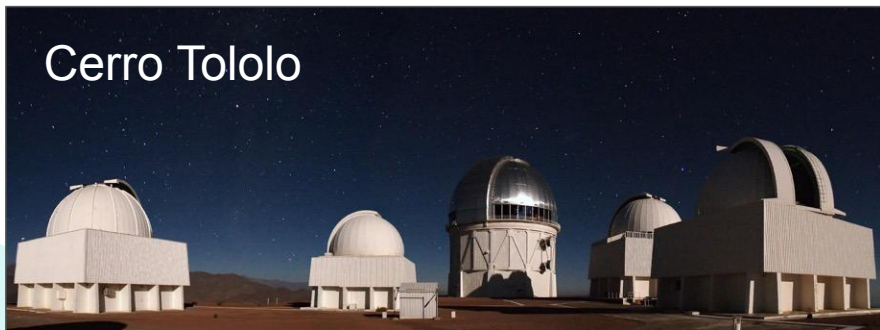


Gemini

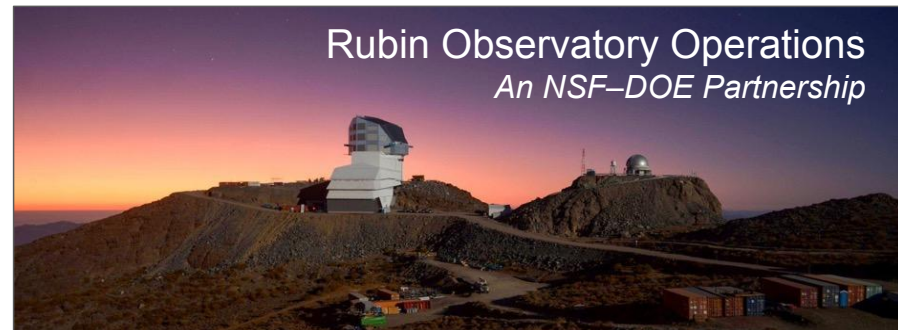
*With our international
partners*



Cerro Tololo



Rubin Observatory Operations
An NSF-DOE Partnership





Where we stand

US Leads the World in:

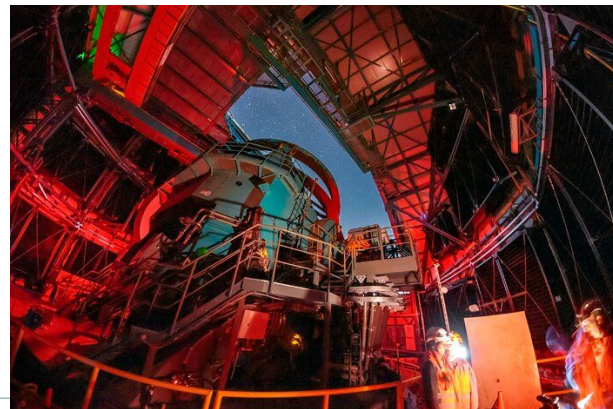
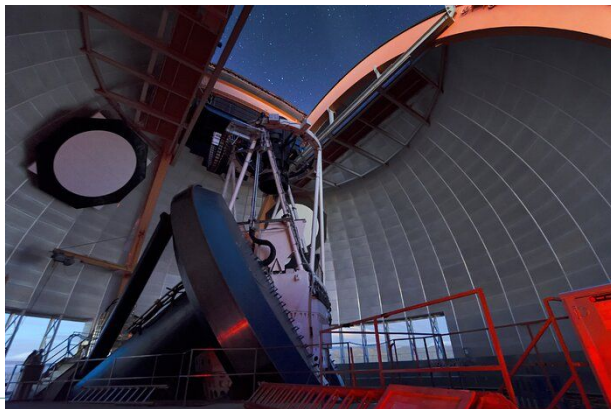
Time Domain Astronomy
High Multiplex Spectroscopy
Wide-Field Surveys & Survey Science

US is Competitive in:

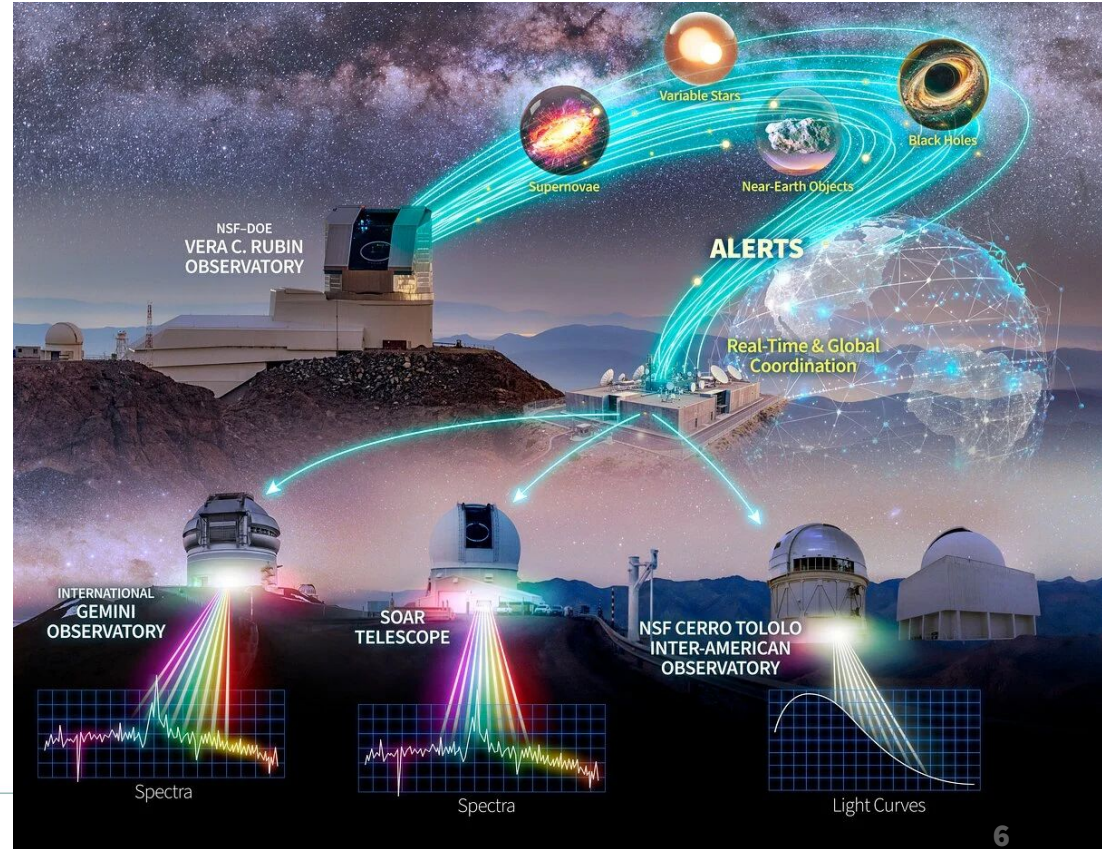
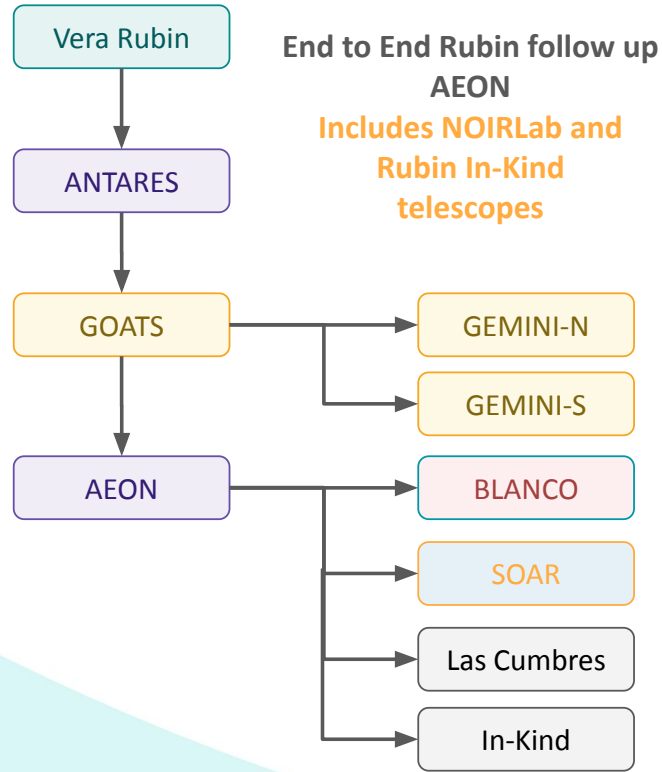
Adaptive Optics
Extremely Large Telescopes
Extreme Precision Radial Velocity

NOIRLab DOE Collaboration, FNAL, LBL, SLAC

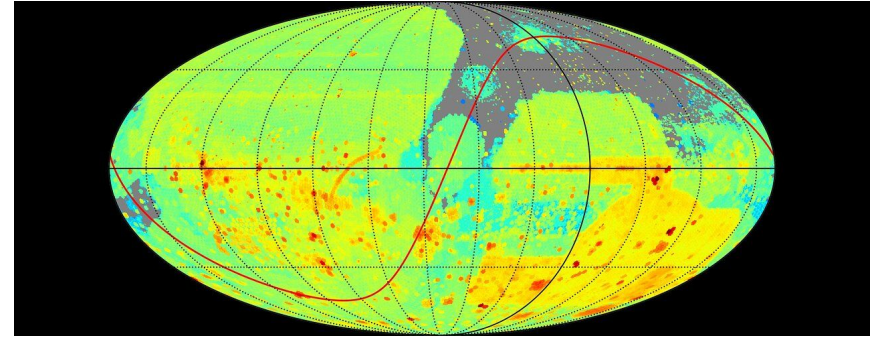
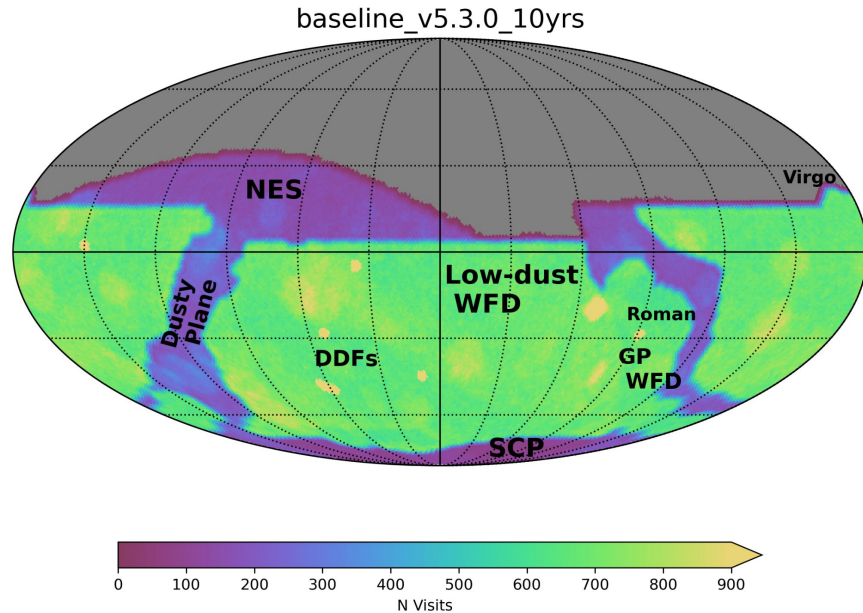
- Dark Energy Camera, DES + Community Surveys (DES, Legacy Survey, DELVE ...)
- Dark Energy Spectroscopic Instrument
- Rubin Observatory and the Legacy Survey of Space and Time
- NOIRLab and its DOE partners have deep and effective relationships that can be leveraged for future science goals.



NOIRLab's immediate future is built around Rubin Observatory as a discovery engine I: Astronomical Event Observatory Network



NOIRLab's immediate future is built around Rubin Observatory as a discovery engine II: Big Data

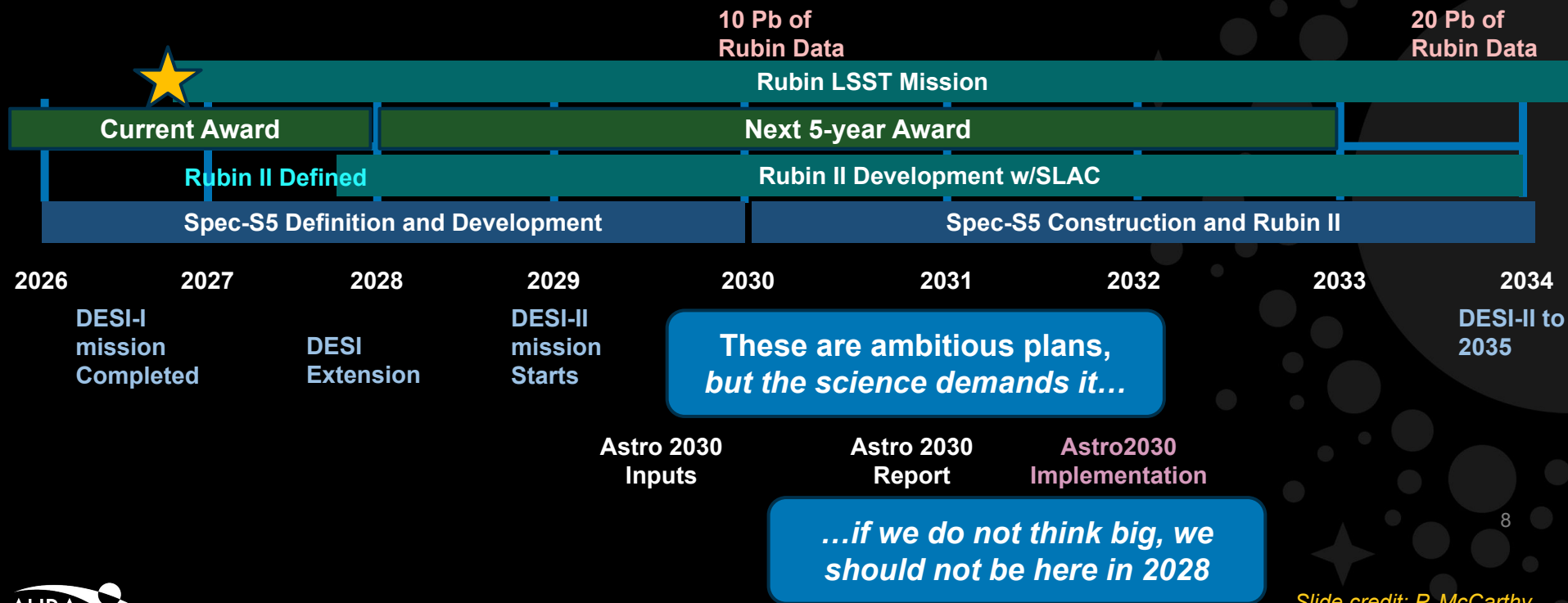


Serving ~2 Pbytes now at the Astro DataLab, imaging and spectroscopic data sets.

NOIRLab has long term interest in archiving and serving the community with data from wide range of instruments and surveys.

Timeline 2026 – 2034

NOIRLab is preparing its next 5 yr proposal to NSF now. Rubin beyond the LSST is a development focus of this proposal. NOIRLab scientists are engaged on Spec S5 as well.

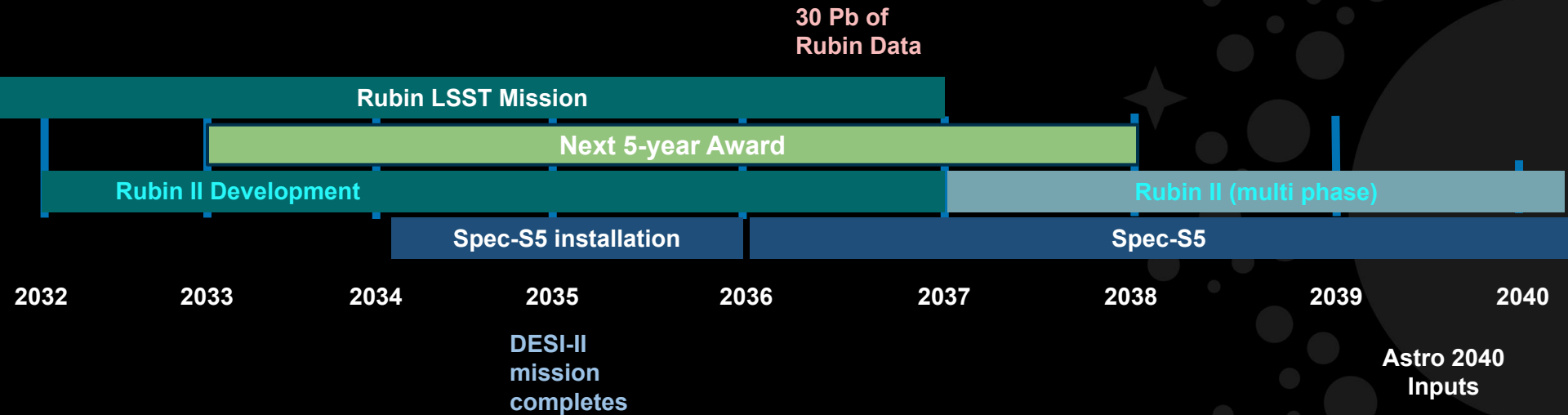




Timeline

2032 – 2040

NOIRLab is preparing its next 5 yr proposal to NSF now. Rubin beyond the LSST is a development focus of this proposal. NOIRLab scientists are engaged on Spec S5 as well.



Programmatic Constraints and Opportunities

- Development timescale is long.
- The LSST clock is ticking.
- We cannot build an entirely new system in less than 10 years.
- But we don't have to.
- We must pursue multiple, phased future programs and development opportunities based on the science aspirations of our combined communities.



Programmatic Constraints and Opportunities

- Future operations and new facilities will inevitably center around more data and processing. Not just about building new hardware.
- Rubin is a system. Data processing and serving data products is at least as costly as running the hardware to capture the cosmos for LSST. What about beyond LSST?
- Our plans must include the continued evolution and development of Rubin as a system.
- Data processing costs must be factored into future plans if we are to be successful. That includes curating the LSST, at least for NOIRLab.
- Federal funding is not likely to grow substantially in the near term, but we must concentrate first on building the science case we want to pursue.
- NOIRLab and SLAC are the Operations partners for the LSST. We should be looking for and open to additional partners going forward.